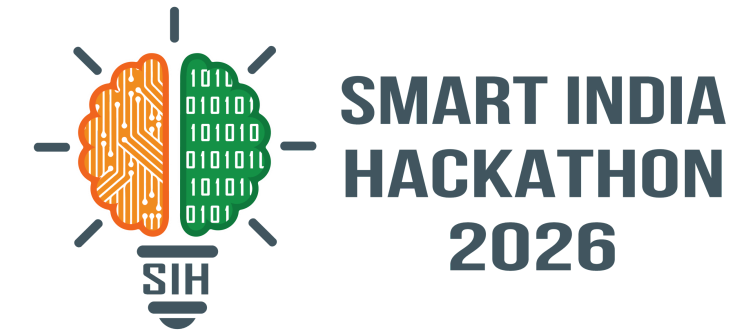
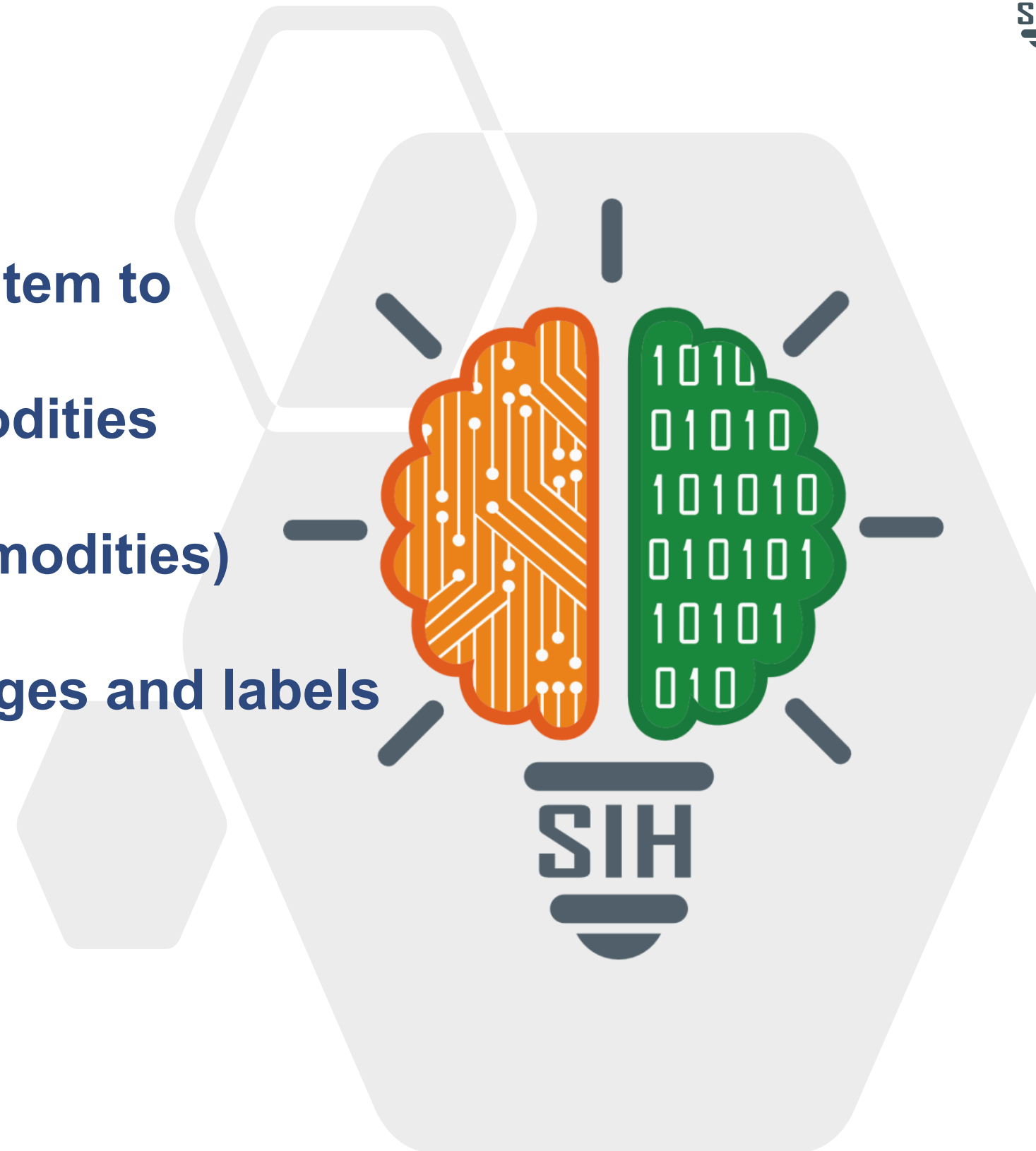


SMART INDIA HACKATHON 2026



- Problem Statement ID - 26034
- Problem Statement Title - Software System to check compliance of Packaged Commodities under Legal Metrology(Packaged Commodities) Rules, 2011 by scanning products, images and labels
- Theme - Miscellaneous
- PS Category- Software
- Team Name - LEGALMATRIX



THE PROBLEM

- Every prepackaged product must carry MRP, net quantity, manufacturing details and consumer care information under Rules 6 to 18 of the LMPC Rules, 2011
- Manual inspection cannot match the scale of physical retail and fast growing e-commerce and quick commerce listings.
- Missing declarations, undersized MRP fonts and incorrect country of origin remain frequent, and origin disclosure is now a named priority under the 2026 amendment for e-commerce imports.

PROPOSED SOLUTION

- Web and mobile application scans a package image, label photo or product listing and extracts every declaration using computer vision and OCR.
- A configurable rule engine checks presence, wording, placement and font size of each declaration against the LMPC Rules, 2011.
- A compliance report is generated in seconds, with evidence photographs, a pass or fail verdict per rule and an overall risk score.

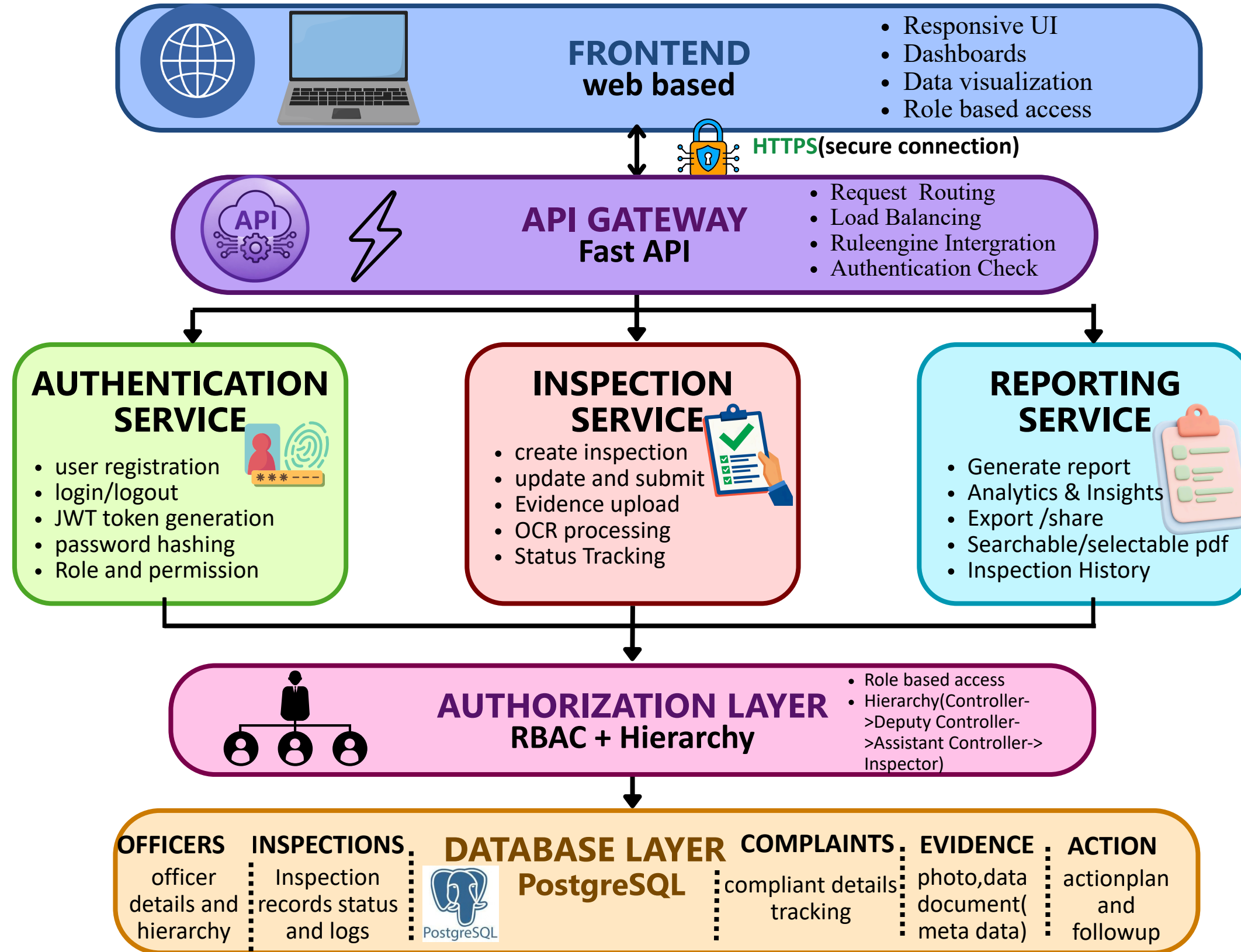
EXPECTED OUTPUT

- Each Legal Metrology officer can screen an order of magnitude more products per inspection day
- Objective, rule traceable scoring reduces disputes and brings consistency across states and inspectors
- A central repository and dashboard feeds the Department of Consumer Affairs and plugs into the eMaap enforcement module

TECHNICAL APPROACH

TECHNOLOGY STACK

-  **MongoDB**
(Caching / Logs)
-  **PostgreSQL**
(Primary Database)
-  **Python 3.x**
(Backend)
-  **FastAPI**
(API Framework)
-  **OpenCV**
(Image Processing)
-  **PaddleOCR**
(OCR Engine)
-  **Docker**
(Containerization)



FEASIBILITY AND VIABILITY

FEASIBILITY

TECHNICAL

- AI-based OCR + image analysis + NLP + commodity-aware rule engine

DATA

- Product-label images + structured declaration fields + Legal Metrology rules

DEPLOYMENT

- Web/mobile application with cloud or department-hosted backend

SCALABILITY

- Expand from selected commodities to multiple product categories and regions

OPERATIONAL

- Officer reviews and verifies AI results before final report

SECURITY

- Role-based access, encrypted storage and audit trail

VIABILITY

GOVERNMENT ADOPTION

- Supports digitization of existing Legal Metrology inspection and reporting workflows.

COST EFFECTIVE

- Uses existing smartphones/cameras and software-based AI/OCR, reducing dependence on specialized hardware.

SECURE AND ACCOUNTABLE

- Role-based access, audit trails, timestamps, GPS and photographic evidence support trustworthy enforcement records.

HUMAN IN LOOP

- Officers can review and verify AI-extracted information before finalizing a compliance report.

LONG TERM VALUE

- Accumulated inspection data enables compliance history, trend analysis and data-driven enforcement.

SUSTAINABLE

- Rules are maintained in an externalized rule repository, allowing updates without rebuilding the application.

CHALLENGES

Challenge	Mitigation
Poor-quality / curved packaging	Image preprocessing + OCR confidence
Different commodity requirements	Commodity-aware rule engine
Changing regulations	Versioned external rule repository
OCR errors	Editable fields + officer verification
No internet in field	Offline-first inspection capture
Evidence integrity	GPS + timestamp + photo + audit trail
Large number of inspections	Central database + analytics dashboard

IMPACT



FEATURE COMPARISON

Existing System vs Our AI-Powered Inspection System

EXISTING SYSTEM

OUR SYSTEM

	Product Image Capture Capture product/package images during inspection	 Limited Manual capture, inconsistent quality	 Fully Supported High-quality image capture with guidance
	AI OCR & Data Extraction Automatically extract data from labels, declarations, documents	 Not Available No automation, manual data entry	 Fully Supported AI OCR extracts accurate data instantly
	Commodity-Aware Rules Apply rules based on commodity type and category	 Not Available One-size-fits-all approach, prone to gaps	 Fully Supported Dynamic rules tailored to commodity and category
	Automated Compliance Check Check compliance against rules and regulations	 Partially Available Manual verification, inconsistent	 Fully Supported Instant automated compliance verification
	Explainable Violations Provide clear reasons for violations	 Not Available Limited visibility, hard to interpret	 Fully Supported AI explains violations with clear reasoning
	GPS + Timestamp Evidence Capture location and time for authentic inspection	 Partially Available Manual logging, not always accurate	 Fully Supported Automatic GPS + timestamp with verified evidence
	Offline Field Inspection Work offline in low/no network areas	 Partially Available Limited offline capability, sync issues	 Fully Supported Seamless offline mode, auto-sync when online
	Digital Report + Analytics Generate digital reports and insights	 Partially Available Basic reports, limited insights	 Fully Supported Comprehensive reports and advanced analytics

BENEFITS

- **Faster Inspections** - Automates label reading, data extraction, and preliminary compliance checks, reducing the time required for manual inspection.
- **Accurate & Consistent Checking** - A standardized rule engine applies relevant compliance checks consistently across different products and inspections.
- **Explainable Violations** - Instead of simply marking a product as non-compliant, the system identifies the **specific failed declaration, condition, and supporting evidence**.



Prototype Layout



SMART INDIA
HACKATHON
2026



[legalmeterology](#)



[DEMO VEDIO](#)

LM

GOVERNMENT OF TAMIL NADU · CONSUMER AFFAIRS

Legal Metrology Enforcement Platform

Packaged commodity compliance verification, field inspection and enforcement oversight platform.

- Citizen Mobile Verification
- Report a problem such as overcharging above MRP, missing label declarations, or short quantity.
- 2. Package & OCR Capture
- Mandatory Package Declarations
- Maintain state-wide Legal Metrology enforcement oversight.

Continue to the platform

Select your role to authenticate securely.

English

Citizen

Report packaged-product discrepancies and track complaints.

Inspector of Legal Metrology

Conduct field inspections and verify packaged-commodity declarations.

Deputy Controller of Legal Metrology

Manage enforcement activities within the assigned district.

Assistant Controller of Legal Metrology

Oversee enforcement operations across the assigned region.

Controller of Legal Metrology

Maintain state-wide Legal Metrology enforcement

Chennai District Overview

Enforcement activity for Chennai district.

Field Officers	2	Total Inspections	9
Violations Flagged	2	Compliant	6
Complaints Pending	2	Resolved	0

Enforcement Overview

9 inspections in scope



Compliant — 6
Non-Compliant — 2
Pending — 1

Recent Inspections

ESTABLISHMENT	PRODUCT	INSPECTOR OFFICIAL SIGN-OFF
Sri Lakshmi Stores INS-2026-1001	AquaPure Natural Mineral Water	R. Karthikeyan
Anna Nagar	GoldDrop	

Legal Metrology
— Citizen Portal
DEPT. OF CONSUMER AFFAIRS, TAMIL NADU

Home

Check a Package

Submit Complaint

Track Complaint

Switch Role

English

LEGAL METROLOGY · CITIZEN PORTAL

Check Packaged Product Labels

Verify mandatory declarations on packaged commodity labels (MRP, Net Quantity, Dates, Manufacturer, and Consumer Care) under Legal Metrology (Packaged Commodities) Rules, 2011.

Check a Package



Check a Package

Upload or take a photo of any package label to automatically check required statutory declarations.



Submit a Complaint

Report a problem such as overcharging above MRP, missing label declarations, or short quantity.



Track a Complaint

Check the current status and resolution of your complaint using your reference ID.

29 Sept
2026, 02:33
pm

Compliant

PDF

29 Sept
2026, 02:30
pm

Compliant

PDF

29 Sept
2026, 02:29
pm

Compliant

PDF

29 Sept
2026, 02:24
pm

Compliant

PDF

28 Sept
2026, 02:38
pm

completed

PDF

28 Sept
2026, 12:36
pm

Pending

PDF

Recent Complaints



No complaints assigned

Citizen complaints routed to you will appear here.

ENFORCEMENT ACTIVE

English

GS

G. Sivakum
District Cont

Sri Lakshmi Stores INS-2026-1001	AquaPure Natural Mineral Water	R. Karthikeyan
Anna Nagar Supermarket INS-2026-1002	GoldDrop Refined Sunflower Oil	R. Karthikeyan
Aavin Retail Outlet INS-2026-1003	Clean&Bright Detergent Powder	R. Karthikeyan
Nilgiris Fresh Mart INS-2026-1004	FreshLeaf Herbal Face Wash	R. Karthikeyan
Nilgiris Fresh Mart INS-2026-1005	GoldDrop Refined Sunflower Oil	S. Meenakshi
Ram Traders INS-2026-1006	Clean&Bright Detergent Powder	S. Meenakshi

Recent Complaints

INSPECTION ID	PRODUCT	VERDICT
LM-CMP-2026-10042 28 Aug 2026	AquaPure Natural Mineral Water	Complaint received
LM-CMP-2026-10058	AquaPure Natural	Under Review

LEGAL AND GOVERNMENT SOURCES

- Legal Metrology (Packaged Commodities) Rules, 2011, Department of Consumer Affairs
- Legal Metrology (Packaged Commodities) Amendment Rules, 2026, e-commerce country of origin provisions (JSA / Mondaq legal updates, 2026)
- National Legal Metrology Portal, eMaap, launch press release, PIB, December 2024
- e-Jagriti and AI enabled consumer protection reforms, 2025, Ministry of Consumer Affairs, Food and Public Distribution (IANS, January 2026)
- Legal Metrology FAQs on font size and declaration compliance, azb Partners legal analysis, November 2025

TECHNOLOGY AND INDUSTRY SOURCES

- Automating FDA food label compliance with computer vision and OCR pipelines, Roboflow, January 2026
- AI regulatory agents for CPG label compliance, computer vision architecture, Signify, 2026
- AI powered label inspection, OCR and print quality verification, SoftwebSolutions, August 2025
- AI vision for OCR, code reading and label verification across FDA and EU FIC frameworks, DigitFactory
- India FMCG industry analysis and market size, IBEF.

Official references

- [Department of Consumer Affairs — Legal Metrology](#)
- [Legal Metrology Overview](#)
- [Legal Metrology_\(Packaged Commodities\) Rules, 2011 — Government publication](#)